

FOOD WEBS AND TROPHIC INTERACTIONS IN INTERMITTENT RIVERS AND EPHEMERAL STREAMS

4.7

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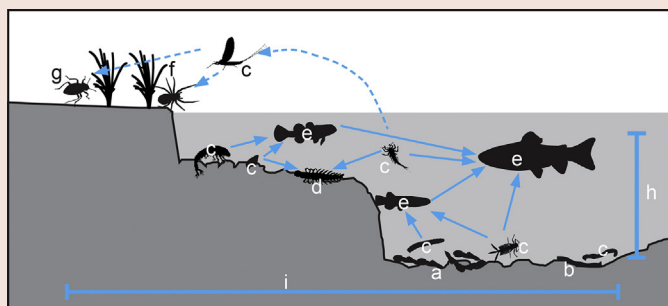
- Contraction of aquatic habitats and expansion of terrestrial habitats associated with cessation of flow and drying in intermittent rivers and ephemeral streams (IRES) change the types, sizes, and abundance of organisms present; the interactions among those organisms; and the overall extent and nature of their trophic interactions
- Drying affects microbial communities and thus detritus processing, heterotrophy can sustain stressed food webs, and autotrophy is important where conditions allow algal buildup
- Spatial and temporal fragmentation of habitats can constrain, compress, and sometimes expand trophic interactions, often substantially increasing interaction strengths as drying progresses and affecting the lateral linking of aquatic and terrestrial food webs
- Overall, drying reduces aquatic food chain length, trophic diversity, and connectance, but also triggers switches toward terrestrial energy pathways with the extent of food web modification linked to regional differences in climate, biota, and riparian conditions

4.7.1 INTRODUCTION

The loss of streamflow, and associated contraction of aquatic ecosystems and expansion of terrestrial ecosystems, inescapably alters the dimensions, conditions, resources, and connections in those habitats (Rolls et al., 2012). This changes the types, sizes, and abundance of organisms present, and the interactions among those organisms, including the extent and nature of their trophic interactions. Thus, the food webs of aquatic and terrestrial systems and their linkage are all affected by the loss of flow and subsequent drying. Here, we review how these linked food webs and trophic interactions are affected by flow cessation and drying, highlighting the mechanisms involved, the emergent patterns within and across river systems, and the insights that can be gleaned both for management and ecological understanding.

BOX 4.7.1 FOOD WEB FUNDAMENTALS IN IRES

Food webs characterize the patterns of energy flow within and between ecosystems by describing the trophic links and interactions among species, including their use of resources (Fig. 4.7.1). At the base of IRES food webs are heterotrophic and autotrophic energy resources. Materials from outside the aquatic ecosystem (i.e., allochthonous sources) including leaves and other organic materials are the major contributors to a detritus pool, which together with the microbes that colonize them (Chapter 4.1), form the heterotrophic energy sources (a). Algae (b) are the major contributors to autotrophic energy pools (Chapter 4.2) and are regarded as an autochthonous energy source because the production is derived within the aquatic ecosystem. Primary consumers (c) of those resources in streams are often classified by the way they feed (Chapter 4.3), and include aquatic invertebrate grazers or scrapers feeding primarily on algae, collector-browsers which feed on a variety of food sources, specialist aquatic detritivores often called shredders which play a key role in breaking down wood and leaf litter (coarse particulate organic matter, CPOM) to finer particles (or fine particulate organic matter, FPOM) which can also be utilized by filter feeders. At higher trophic levels, secondary consumers include a range of predatory aquatic invertebrates (d) and fish (e, Chapter 4.5) in aquatic habitats. Birds, spiders (f), and other predatory taxa (Chapter 4.6) which live in riparian zones can also feed on adult stages of aquatic taxa (dotted arrows), or aquatic stages when they are trapped in pools of drying IRES. After IRES dry, terrestrial detritivores (g) are often important consumers of decaying aquatic organisms and other detritus (Chapter 4.4). Collectively these interactions can be characterized by the length of food chains, a measure of trophic height (h), and the variety of trophic interactions or trophic width (i).

**FIG. 4.7.1**

Concepts related to IRES food webs and interactions in aquatic (solid arrows) and terrestrial (dashed arrows) environments, including energy sources, primary and secondary consumers, and trophic dimensions. Abbreviations are as follows: a, heterotrophic sources; b, algae; c, aquatic primary consumers; d, predatory aquatic invertebrate; e, fish; f, spider; g, terrestrial detritivore; h, trophic height; i, trophic width. Some silhouettes and original concept developed by Amanda Klemmer, and some original silhouettes also courtesy of the Integration and Application Network, University of Maryland Center for Environmental Science (ian.umces.edu/symbols/).

Food webs (Box 4.7.1) reflect emergent patterns that can be resolved from the feeding interactions of organisms, providing a powerful way of evaluating the structure and functioning of an ecosystem (Thompson et al., 2012). Moreover, because food webs reflect connectivity between organisms and resources (Fig. 4.7.1), they are useful for understanding and summarizing the highly dynamic changes that intermittent rivers and ephemeral streams (IRES) undergo during wetting and drying cycles.

Much of our understanding about how ecosystems maintain structure, are assembled, and function has been based on the patterns that are apparent in food webs. Thus, the process of food web disassembly/assembly (Fig. 4.7.2) associated with drying and rewetting reveals processes and mechanisms relevant to many ecosystems. In particular, theory highlights the importance of habitat size (McCann et al., 2005), the presence of compartments encompassing the resources associated with particular prey

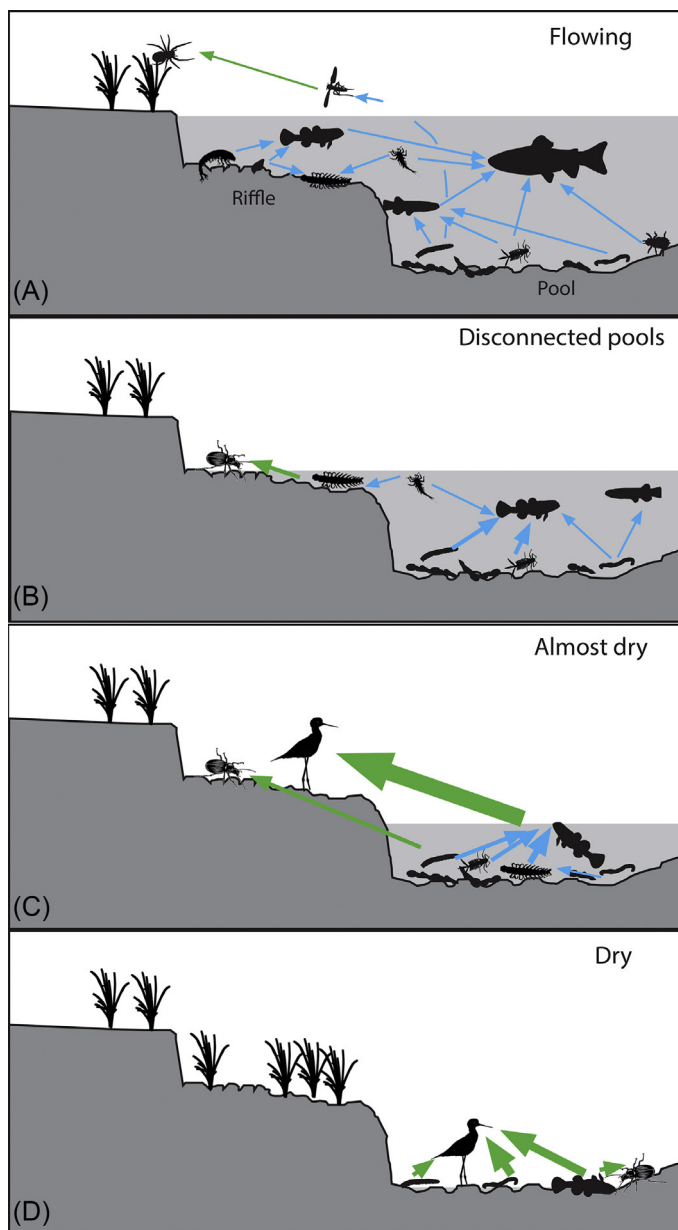


FIG. 4.7.2

Changes in aquatic (*blue arrows*) and terrestrial (*green arrows*) trophic interaction strength (arrow width) associated with flow intermittence (increasing from A to D), illustrating typical changes in interactions and food webs. These involve reductions in body size and richness of consumers, but increased density and interaction strength, associated with shrinking aquatic habitat dimensions over the hydrological cycle of IRES. There is also an associated expansion of terrestrial habitats, with terrestrial predators often feeding on dense aggregations of aquatic prey trapped in fragmented aquatic habitats. Some silhouettes and original concept developed by Amanda Klemmer, and some original silhouettes also courtesy of the Integration and Application Network, University of Maryland Center for Environmental Science (ian.umces.edu/symbols/).

or energy types, the prevalence of particular interactions (or modules) in food webs, and the distributions of both body sizes and interaction strengths (McCann et al., 1998; Thompson et al., 2012). As all of these may be strongly affected by the wetting and drying cycles characteristic of many IRES, much is to be gained from investigating IRES food webs.

The utility of insights derived from studying the food webs of IRES depends, in part, on the consistency with which processes are defined and then used by ecologists and practitioners. Here we are concerned with actual IRES rather than experimental channels, but we briefly summarize insights gained from the latter. We use standard terminology (after Rolls et al., 2012) to refer to IRES throughout, unless giving examples of a particular system, flow regime, or hydrological phase. We first provide an overview of processes affecting IRES food webs in the broadest sense. Then, working across trophic levels, we describe interactions at the base of food webs focusing on how changes in autotrophy and heterotrophy with drying alter resource availability. Next, we consider higher trophic levels, focusing on mechanisms affecting consumer-resource interactions and the spatial dimensions over which changes occur across hydrological phases. We highlight how regional variations in hydrology, vegetation, biodiversity, and aspects of local context affect food webs. Following this, we describe changes in high-level food web properties (e.g., food chain length) across hydrological phases and finish with a look to the future, considering how changes in habitat size, temperature, and water quality might influence food webs.

4.7.2 THE BASE OF IRES FOOD WEBS

The source, quantity, and lability of organic matter can significantly influence trophic interactions from the bottom up. Understanding controls on and the contribution of terrestrial sources of detritus and organic matter processing (i.e., heterotrophy) versus photosynthetically fixed sources of carbon (i.e., autotrophy) is important for understanding food webs in all rivers (Hall and Meyer, 1998). The contribution of detritus to food webs has long been recognized (Hall et al., 2000; Schmid and Schmid-Araya, 2010). Macroinvertebrate shredding is the primary physical breakdown pathway for leaf litter and other coarse organic matter sources (Chapter 4.3) whereas microbial communities release enzymes to enable further breakdown of organic matter compounds (Chapter 3.2). Organic matter from both terrestrial and instream sources make up the majority of invertebrate community diets (Reid et al., 2008; Dekar et al., 2009), increasingly so in times of low flow in some (Closs and Lake, 1994), but not all cases (Álvarez and Pardo, 2009; Vannucchi et al., 2013).

Hydrological extremes alter detrital pools in IRES by altering the quantity and quality of organic matter and the microbial communities responsible for detrital processing (Romaní et al., 2012; Chapter 3.2). During drying, surface water becomes physically contracted (Chapter 2.3) and increasingly disconnected from terrestrial inputs, nutrient processing becomes spatially patchy (Bernal et al., 2013), and food webs become smaller (McHugh et al., 2015). This causes leaf litter breakdown rates to vary as a function of macroinvertebrate shredder density, microbial community and activity rates, and the degree of immersion in water (Bruder et al., 2011; Corti et al., 2011). Microbial community structure itself may drive certain ecosystem functions over others. Sediment experiments suggest that microbial communities responsible for organic matter processing are both altered and reduce their functioning upon drying (Amalfitano et al., 2008; Chapter 3.1). Microbial respiration rates across an intermittent-perennial stream transition indicated carbon stocks to be between four- and eight-fold higher in perennial streams than in either IRES or terrestrial sites (Gerull et al., 2011). Thus, drying likely alters food availability at the food web base. For example, in Mediterranean IRES, the lack of

shredders has been associated with the poor quality of organic matter after flow returns, and scrapers or collector-browsers feeding on autochthonous food sources are the dominant consumers in these systems (Álvarez and Pardo, 2009; López-Rodríguez et al., 2012; Vannucchi et al., 2013).

Altered functioning and food web structure may be influenced by shifts in microbial assemblages in longitudinal, vertical, and lateral dimensions during different hydrological phases (Chapter 4.1). Febria et al. (2015) used network path analysis to compare microbial groups in a headwater stream network in Chesapeake Bay, USA, and found perennial streams were dominated by carbon fixers and IRES by nitrogen fixers. The pattern became more enhanced during periods of drying. However, the differences were not attributed to hydrological regime *per se* but to soil moisture, a physical attribute associated with specific streambed microhabitats. Overall evidence suggests that different microbial taxa and functions are associated with IRES food webs compared to perennial streams.

The primary producers in IRES are mostly algae, with macrophytes playing a relatively minor role (Chapter 4.2). Primary production can be considerable in some IRES, such as desert streams. For example, in spatially intermittent Sycamore Creek (Arizona), Fisher et al. (1982) recorded high primary production and rapid recovery of algal communities in response to flooding. The creek's algal assemblage was generally dominated by disturbance-resistant diatoms, but also included macroalgae and cyanobacterial mats (Grimm and Fisher, 1989). Álvarez and Pardo (2009), working in Mediterranean Spain, reported substantial changes in algal abundance measured by chlorophyll pigments over the course of a year. These studies illustrate that primary production can be highly variable in IRES, but is particularly high in warm, open-canopied systems when flows are conducive.

Assessments of how temporal changes in hydrological regime—whether they be short term (e.g., days, seasons) or long term (e.g., years, decades)—show further evidence of shifts between autotrophy and heterotrophy (Fig. 4.7.3). Reid et al. (2008) working in IRES in southeast Australia reported seasonal shifts in basal resources from primarily heterotrophic year-round to autotrophy in some

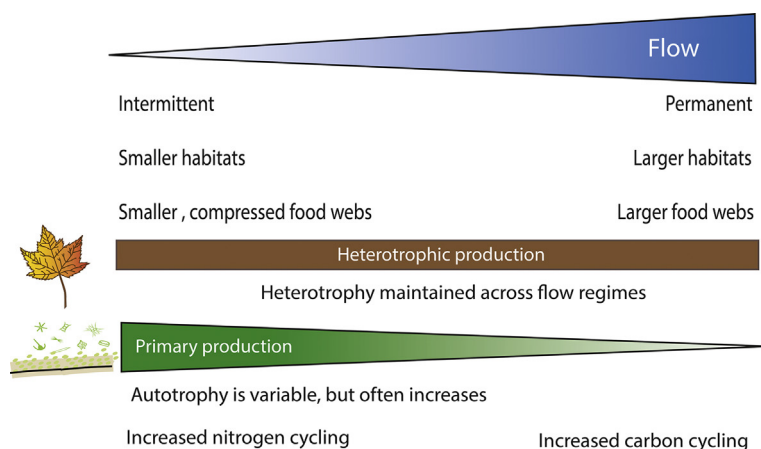


FIG. 4.7.3

Changes in basal food web components across hydrological phases (flow = blue) of IRES, indicating constant heterotrophy (brown) across flow states, increased autotrophy (green) with drying, and changes in the relative increases in nitrogen and carbon cycling. Leaf and algal symbols courtesy of the Integration and Application Network, University of Maryland Center for Environmental Science (ian.umces.edu/symbols/).

streams during periods of disconnection in spring and summer. Across multiple studies, terrestrial carbon pools are a reliable energy source in many IRES (Closs and Lake, 1994; Reid et al., 2008; Dekar et al., 2009; Courtwright and May, 2013), with shifts toward autochthonous production in pools during drying (Fig. 4.7.3). Studies from Mediterranean Europe point toward reliance on autochthonous resources during drying (Álvarez and Pardo, 2009; Vannucchi et al., 2013). Thus, it appears that heterotrophy generally dominates during wet phases, but algal resources and autotrophy can assume prominence during drying/low-flow periods. Further investigation into these patterns at a range of scales is warranted, particularly for determining which carbon source(s) matter most to production at higher levels, whether source shifts are linked to changes in the microbial communities (i.e., structure, functions) and the extent to which ongoing climate warming may impact these food sources.

4.7.3 CONSUMER-RESOURCE DYNAMICS IN IRES

The variable nature of IRES also structures the exchange of matter and energy among trophic levels. Hydrological contraction and fragmentation during drying cause aquatic taxa to concentrate in wetted areas (Fig. 4.7.4). Once fragmentation has occurred, density-dependent processes influence trophic structure and functioning. Densities of predators increase during drying as pools shrink and water quality deteriorates (Dekar and Magoulick, 2007; Dewson et al., 2007; Walters and Post, 2011), and predator-prey ratios typically increase as well (Bogan and Lytle, 2007; McHugh et al., 2015). Strong biotic interactions are usually accompanied by equally strong abiotic stress associated with deteriorating physicochemical conditions (Boulton and Lake, 1990; Drummond et al., 2015; McHugh et al., 2015). Then, upon rewetting, there can be periods when primary producers or primary consumers escape top-down control until herbivores or predators recolonize (Fig. 4.7.4b).

Spatial variation of trophic interactions may be less marked than temporal variability (Closs and Lake, 1994), but the occurrence of strong interactions generally increases with drying, along with changes to abiotic factors that stress aquatic taxa (Stanley et al., 1997). The distribution and abundance of predators depend on the timing and intensity of drying, barriers to movement, and species-specific environmental tolerances and mobility (Walters, 2016). These factors determine the spatial variation in ecological communities, likely causing changes in predation intensity at small spatial scales (Schlosser and Ebel, 1989; Acuña et al., 2005). Pool size and depth are especially important in IRES, determining the density and composition of animal communities (Englund and Krupa, 2000; Magoulick and Kobza, 2003; Wesner, 2013). Abiotic factors may be more important than biotic interactions in shaping fish assemblages of drying stream pools (Capone and Kushlan, 1991; Magoulick, 2000), although interactions can be intense (Power et al., 1985; Williams et al., 2003) and predation by terrestrial vertebrates may be understated (Capone and Kushlan, 1991; Magoulick, 2000).

Hydrological alteration associated with drying may also mediate biotic interactions with nonnative species and affect invasion outcomes. Nonnative fishes often originate in lentic habitats and are less flood-adapted than native species (Propst and Gido, 2004; Kiernan et al., 2012; Murphy et al., 2015), so habitat drying may alter invasion likelihood. However, native IRES fauna is often well adapted to drying (e.g., physiological tolerance, Labbe and Fausch, 2000; Magoulick and Kobza, 2003), so the net results on interactions between native and introduced species are unclear (Propst et al., 2008). Drying has been shown to favor native galaxiids over invasive brown trout in Australia and New Zealand (Closs and Lake, 1996; Leprieur et al., 2006). In other cases, drying may facilitate invasions. Drying has favored

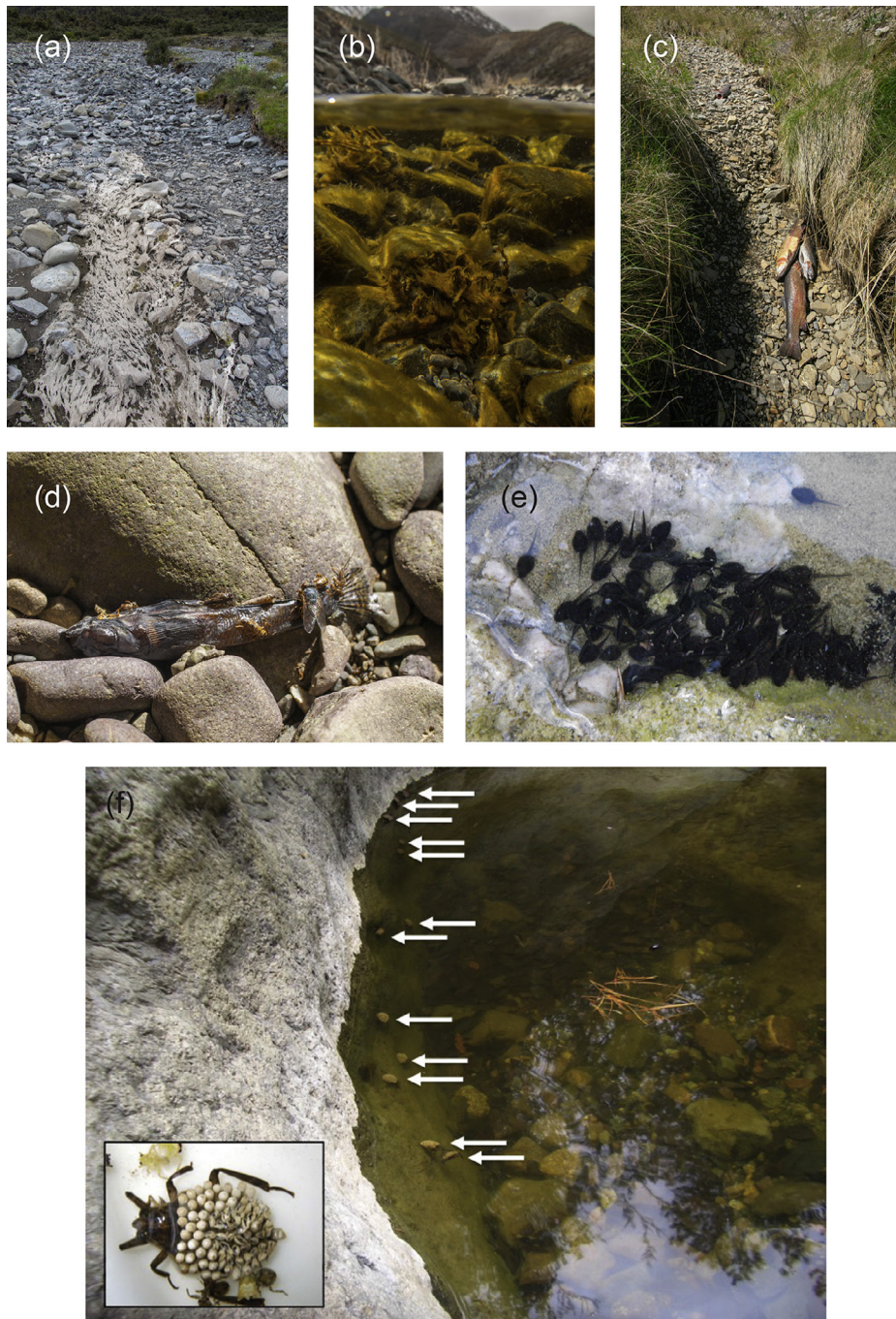


FIG. 4.7.4

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the displacement of native riparian trees by invasive *Tamarix ramosissima* in southwestern United States (Stromberg et al., 2007). In Arkansas and Missouri, the native crayfish *Orconectes eupunctus* has been replaced by the invasive *Orconectes neglectus* (Larson et al., 2009; but see Magoulick, 2014). In Spain, invasive fishes (e.g., *Lepomis gibbosus*) seem favored by drying compared to endemic cyprinids (Bernardo et al., 2003). Such findings indicate context- or condition-dependency, a pervasive theme in IRES ecology (Walters, 2016), suggesting a match between hydrology and life history mediates native-invader interactions. Thus, understanding the differential response of species to drying not only informs the study of IRES food webs but is also crucial for conservation and mitigating the impacts of climate change and flow alteration (Chapter 5.4).

The large size and limited dispersal ability of many aquatic predators make them especially vulnerable to effects of spatial fragmentation. During drying, they can become confined in pools (Dekar and Magoulick, 2007) and risk extirpation if pools dry completely (Bogan and Lytle, 2011; Christian and Adams, 2014). Upon rewetting, recolonization by large predators is dependent upon dispersal ability (Boersma et al., 2014; Cañedo-Argüelles et al., 2015). Bogan and Lytle (2011), for example, recorded the transition from a perennial but fragmented (isolated pools) state to one of complete drying and noted the loss and prolonged absence of large, dispersal-restricted, predators and their replacement by smaller vagile mesopredators in a multiyear study in Arizona. The variable hydrology of IRES (Chapter 2.2) may increase the vulnerability of IRES fauna to unexpected or prolonged drying and anthropogenic water abstraction (Deacon et al., 2007) and suggests hysteresis effects are likely.

Transitions between lotic and lentic phases generate turnover in aquatic communities, driven frequently by predator assemblages. Bogan and Lytle (2007) highlighted this pattern in Arizonan IRES, where they observed rapid turnover in aquatic invertebrate communities between high- and low-flow seasons. They studied several streams and recorded an increase in the proportion of predators from 24.5% during high flow to 75.2% during low-flow seasons. Large-bodied invertebrate predators (Belostomatidae, Dytiscidae, and Notonectidae) present in low abundances during high flow often become numerically dominant during low flow (Acuña et al., 2005; Bogan and Lytle, 2007).

Even when consumer assemblages remain unaltered between seasons, marked changes in predator diet can occur. In a large floodplain IRES (Cooper Creek, central Australia), Balcombe et al. (2015) found that predatory fishes switched from diets of zooplankton during residency in dry-season pools to a diverse diet of midges, beetle larvae, zooplankton, and terrestrial materials during the wet season spent on the floodplain. Overall, this change led to wider diet breadths for fish collected during wet compared to dry periods (Balcombe et al., 2005). Thus, large changes in trophic interactions can occur across IRES hydrological phases.

FIG. 4.7.4

Variation in resource states and consumer densities across hydrological phases in IRES: (a) algal mats after drying, (b) algal bloom after flow resumption, (c) carcasses of a large number of trout that were trapped in a small pool, (d) flies (a terrestrial invertebrate) on the dried carcass of a fish, (e) tadpoles trapped at high densities in the last remnants of a pool, and (f) *Abedus herberti* (Hemiptera: Belostomatidae) giant water bugs cluster along a 1-m portion of bedrock in a drying pool in Arizona. The giant water bugs are top predators in seasonally fragmenting IRES in southwestern United States and can reach densities of up to 50 m⁻² during stream contraction and drying. The adult males in this photo bear eggs on their backs (inset), which will soon hatch and introduce hundreds of additional juvenile predators to the small pool.

Photos: Courtesy A.R. McIntosh (a–d) and K.S. Boersma (e, f).

Transition from wet to dry phases also affects riparian consumers. McCluney and Sabo (2009, 2012) manipulated water availability in the San Pedro River (Arizona) and recorded the response of riparian arthropod food webs. Interaction strengths between predators (wolf spiders: *Hogna antelucana*) and prey (crickets: *Gryllus alogus*) decreased during dry phases relative to wet phases (McCluney and Sabo, 2009). Greenwood and McIntosh (2010) also documented significant drying-associated changes in populations of riparian fishing spider *Dolomedes aquaticus* in New Zealand. *Dolomedes* biomass was lower and individuals were smaller in drying reaches, reflecting their heavy reliance on aquatic prey and vulnerability to desiccation. Lateral interactions between aquatic and terrestrial habitats are thus important determinants of trophic processes in IRES (Section 4.7.4).

While many taxa that inhabit IRES have adapted to withstand seasonal fluctuations in water availability, dispersal-limited predators such as fish and large invertebrates are still vulnerable to complete drying. The giant water bug *Abedus herberti*, the top predator in many fragmented streams in southwestern United States, is exemplary here—a flightless aquatic invertebrate with limited dispersal capacity and high risk of local extinction due to droughts (Bogan and Lytle, 2011). Researchers measured the consequences of such extinctions to food webs by experimentally removing *A. herberti* from some mesocosms but not others (Boersma et al., 2014). Removal caused an increase in the diversity and abundance of mesopredators and unexpected losses in detritivorous taxa. Instream experiments provide additional evidence of the importance of top-down processes in IRES (Williams et al., 2003; Rodríguez-Lozano et al., 2015) and highlight the importance of predatory fish in fragmented pools during drying. Overall though, little is known about how interannual environmental variability modulates predation in IRES food webs, although shifts in timing and magnitude of flow intermittence are expected to alter proximate abiotic drivers.

4.7.4 MULTIDIMENSIONAL INTERACTIONS WITHIN IRES

Materials and organisms cross ecosystem boundaries and transition zones (Strayer et al., 2003). IRES comprise mosaics of terrestrial and aquatic habitat that transition in space and time (Datry et al., 2016), longitudinally along upstream-downstream continua (Leigh et al., 2010a), laterally across river-riparian zone-floodplain continua (Bunn et al., 2006a), and even vertically among channel beds, subsurface sediments, and groundwater compartments (Stubbington et al., 2011) as habitats expand and contract, and dry and wet with changes in streamflow (Stanley et al., 1997; Chapters 2.3 and 4.9). This shifting habitat mosaic not only influences the movements, stability, and persistence of biota inhabiting IRES, but also the spatiotemporal dynamics of trophic links among them (Polis et al., 1996) and the biota and food webs of adjacent aquatic and terrestrial ecosystems (Fig. 4.7.5).

LONGITUDINAL INTERACTIONS

Waters that travel downstream along IRES channels often do so after dry phases when instream aquatic habitat has been absent or restricted to isolated pools and other moist refuges interspersed between exposed, dry riverbed (examples in Chapter 4.4). These events are known as first pulses or advancing wetted fronts and can entrain substantial amounts of organic material: aquatic invertebrates and fish, terrestrial invertebrates, fine detritus, biofilms, leaves, and wood. Travelling down a 7-km dry reach of the intermittent Albarine River, France, one such pulse contained several orders of magnitude more invertebrates than found in peak-flow pulses of perennial rivers (Corti and Datry, 2012). Terrestrial and

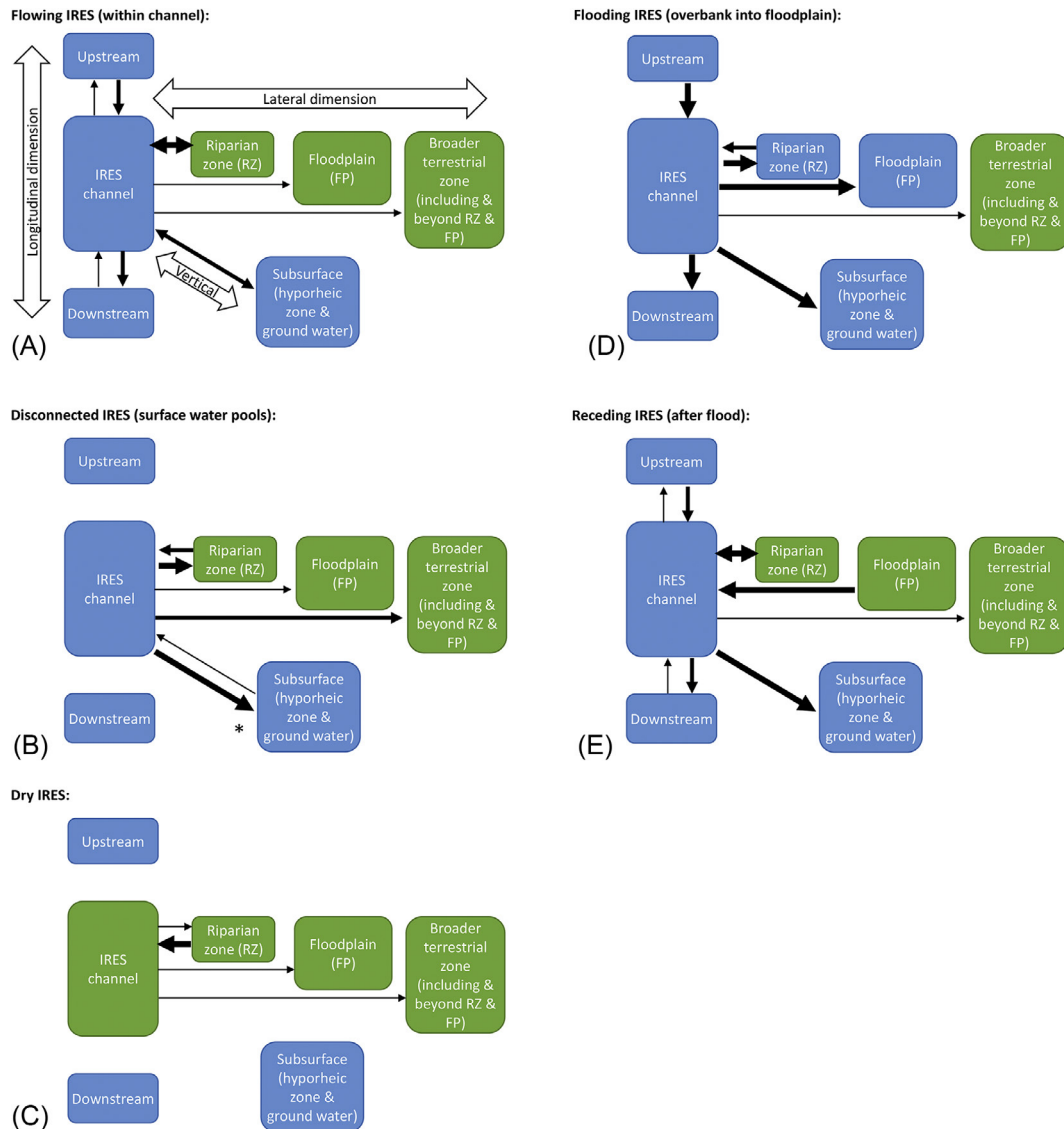


FIG. 4.7.5

The four-dimensional nature of food webs in IRES, showing links between aquatic (blue boxes) and terrestrial (green boxes) compartments along longitudinal (upstream and downstream of the IRES channel), lateral (IRES channel to the riparian zone, floodplain and beyond), and vertical (IRES channel to hyporheic zone and groundwater) dimensions and at different hydrological (temporal) phases: (A) flowing, (B) disconnected pool, (C) dry channel, (D) flooding, and (E) flood recession. Arrows show direction of links between compartments; for example, via the passive or active movement of organisms to another compartment or via consumption of organisms by a predator from another compartment (e.g., by migratory bird species in the broader terrestrial zone consuming aquatic prey in IRES channels), and line widths reflect link strength. Upstream and downstream aquatic compartments can include perennial rivers and streams and IRES. The link indicated with “*” is particularly strong when water level is receding. See text for more details.

aquatic invertebrates along with fine suspended organic material were transported downstream with the pulse, entrained for an average of 2 km before being redeposited on the newly rewetted bed. This represents a substantial transfer of energy and associated consumers (Fig. 4.7.5).

Depending on the size of the pulse and subsequent flows, redeposited material can accumulate in-stream, along margins or even above the active channel on banks or floodplain terraces. The importance of these accumulations to adjacent terrestrial ecosystems and downstream aquatic food webs is conjectured as high (Rosado et al., 2015) but remains understudied. Flow pulses may provide peak supplies of nutrients and food resources that stimulate downstream productivity and encourage fish and other mobile consumers to capitalize on the newly available resources (Corti and Datry, 2012; Rosado et al., 2015). The pulsed supplies may also have negative consequences for downstream reaches. Temporary hypoxia may ensue, resulting in fish kills and loss of other aquatic biota with unknown but potentially negative effects on food web structure (e.g., due to declines in prey) and ecosystem processes (e.g., due to unusually high numbers of dead fauna; King et al., 2012). Low-magnitude pulses in dryland IRES may disrupt the production of algae in shallow littoral zones that sustain aquatic food webs through the extended dry phases that occur between flood pulses (Bunn et al., 2006a). Overall, however, the rewetting of dry channels and longitudinal reconnection of isolated aquatic habitats within IRES networks are important for maintaining the integrity of downstream aquatic ecosystems (Bunn et al., 2006a; Leigh et al., 2010a) and supplying downstream communities with labile organic material for assimilation into their food web architecture (Jacobson et al., 2000).

LATERAL INTERACTIONS

As alluded to earlier, rivers, riparian zones, and floodplains are linked laterally via food webs (e.g., Baxter et al., 2005). In IRES, these trophic interactions occur as in perennial rivers; however, spatial and temporal fluctuation in the presence of water may modify the direction, timing, and intensity of energy flows between the adjacent ecosystems.

Given the dynamic nature of IRES hydrology, subsidies of aquatic resources to riparian food webs are spatially variable and temporally pulsed (Fig. 4.7.5). Contracting aquatic habitats offer a temporarily available source of concentrated, trapped, and often highly visible prey that can subsidize riparian food webs (Section 4.7.3). Dekar and Magoulick (2013) demonstrated predation by birds and mammals on fish trapped in artificial pools designed to mimic disconnected aquatic habitats in IRES, with large-bodied individuals being particularly susceptible to predators. Leigh et al. (2013a) found that up to 50% of the vertebrate fauna observed in riparian zones of isolated waterholes along stretches of otherwise dry IRES in Australia consumed aquatic fauna and at a higher rate than in the riparian zone of a nearby perennial channel. Dead aquatic biota accumulated on recently dried IRES channels can also constitute a pulsed resource subsidy to riparian food webs (Boulton and Suter, 1986; Steward, 2012).

The pulsing of aquatic prey availability not only affects lateral food web connections but also population and community dynamics within riparian zones. For example, the low biomass of aquatic insects in drying reaches of a New Zealand IRES was postulated to contribute to the low biomass of *Dolomedes* spiders (Greenwood and McIntosh, 2010; Section 4.7.3). Aquatic subsidies from Sycamore Creek also explain the high abundance and richness of spiders in riparian areas (Sanzone et al., 2003). Conversely, temporal variation in the abundance of riparian consumers potentially influences aquatic prey populations. In the Tagliamento River, a large braided IRES in Italy (Tockner et al., 2003), seasonal variation in the abundance of riparian predatory arthropods coincides with their proportional use

of aquatic subsidies (Paetzold et al., 2005). Such predation could represent quantitatively important, pulsed transfers of aquatic secondary production to the riparian food web (Paetzold et al., 2005) while providing a further constraint on aquatic prey abundances during seasonal flow extremes.

Terrestrial invertebrates are also an important food resource for IRES inhabitants. Native brook trout inhabiting forested IRES in Virginia, USA, responded negatively to experimental reductions in terrestrial invertebrate inputs (Courtwright and May, 2013). The experimental reduction was mirrored by a proportional dietary response and fish were unable to make up the difference with more aquatic invertebrates.

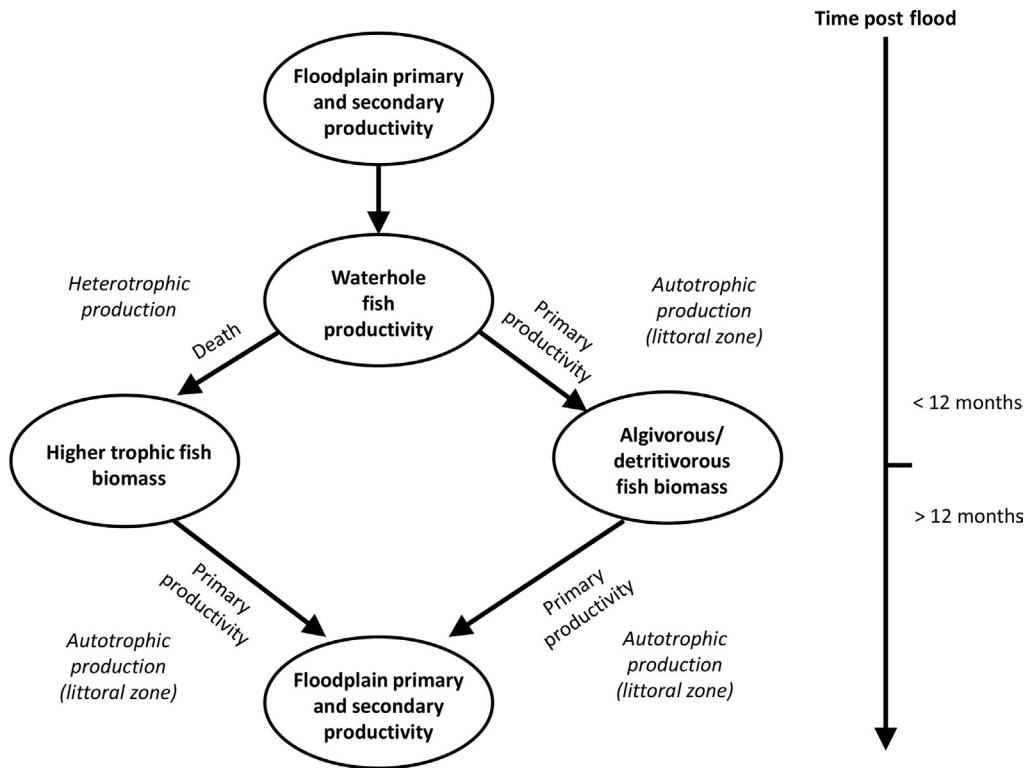
Food web connections between adjacent terrestrial and aquatic ecosystems will depend somewhat on the characteristics of the individual ecosystems (e.g., forested vs. grassland) and regional climate and hydrology (Section 4.7.5). Not all IRES food webs rely on terrestrial subsidies (e.g., Reid et al., 2008; Blanchette et al., 2014), and similarly not all adjacent riparian food webs rely foremost on aquatic production (Corti and Datry, 2014; Sections 4.7.2 and 4.7.3). Beyond riparian zones, lateral interactions between IRES channels and adjacent ecosystems may extend to the floodplain (Fig. 4.7.5; e.g., Sheldon et al., 2002; Tockner et al., 2003). Connections between the in-channel and floodplain compartments in these systems are influenced by the lateral advance and retreat of floodwaters, with the pulsed contraction and expansion of aquatic habitats producing fluctuations in the sources of nutrients fuelling production (e.g., Fig. 4.7.4).

During floods, production booms on dryland IRES floodplains, providing an immense food resource for fish and other aquatic consumers (Bunn et al., 2006a) which sustain the aquatic food web well after floodwaters recede and isolated waterholes reform (Balcombe et al., 2015; Fig. 4.7.6). Burford et al. (2008) estimated that 50% of the fish carbon in a waterhole of Cooper Creek, isolated after floodwaters had retreated, was derived from floodplain food sources. Carbon from fish that boomed at high flows but later died in a waterhole fuelled a temporary boom in the waterhole's heterotrophic production. Between floods during extended dry phases, algal production in the littoral zones of isolated waterholes provides a major source of carbon and nitrogen to both primary and secondary aquatic consumers (Bunn et al., 2003, 2006a). The colonization of dry riverbeds by terrestrial plants, and the expansion of terrestrial ecosystems in general, are also likely to have ramifications for aquatic–terrestrial food web dynamics (Fig. 4.7.6).

Lateral connections between IRES and other ecosystems can be spatially extensive, beyond riparian zones and even floodplains, particularly in regions tending toward aridity with infrequent but major flooding (Fig. 4.7.5). For example, waterbirds respond to massive flooding events in dryland IRES to exploit temporarily productive feeding and breeding habitats (Chapter 4.6; Kingsford et al., 1999). Many bird species with wide foraging ranges and long seasonal migration routes observed near IRES waterholes in Australia's wet-dry tropics are known consumers of aquatic fauna (Leigh et al., 2013a). Subsidies from IRES may therefore extend far beyond waterbodies of origin, highlighting the potentially great, but undocumented, importance of IRES at regional scales.

VERTICAL INTERACTIONS

Variable groundwater inputs alter the duration of flow or water permanence in the surface and hyporheic zones of IRES (e.g., Larned et al., 2008; Chapter 2.3), which can influence ecosystem processes affecting overall food web dynamics, including leaf decomposition and nutrient cycling (Fig. 4.7.5; Boulton and Hancock, 2006; Chapters 2.3 and 3.2). Hyporheic sediments in IRES can remain biologically active

**FIG. 4.7.6**

Processes driving boom-and-bust fish production in the waterholes of Cooper Creek, which flows through the arid regions of central Australia, over the hydrological cycle starting with flooding of the floodplain and followed by flood recession, early phases of disconnection (all <12 months after flooding), later phases of disconnection and eventual drying (both >12 months after flooding). Floodplain flooding triggers very high productivity leading to very high fish biomass as the waters recede, driven by allochthonous resources (the “boom”). As waterholes dry down and disconnect, there is significant fish mortality (Death) leading to increased heterotrophic production due to the microbial breakdown of the floodplain-derived fish biomass together with increasing autotrophic production from littoral-zone algae. With extended disconnection and severe drying (>12 months), waterholes move into the “bust” phase where the minimal rates of production that prevail are maintained by primary producers.

From Balcombe, S.R., Turschwell, M.P., Arthington, A.H., Fellows, C.S., 2015. Is fish biomass in dryland river waterholes fuelled by benthic primary production after major overland flooding? *J. Arid Environ.* 116, 71–76, with permission.

throughout periods of surface-flow cessation due to groundwater connection (Leigh et al., 2013a); yet lengthy dry phases put hyporheic inhabitants under increasing water stress (Datry, 2012; Chapter 4.3). In Fuirosos River, a Mediterranean IRES, heterotrophs in biofilms were more resistant to drying than autotrophs, particularly in the hyporheic zone which was buffered against the water stress caused by surficial drying (Timoner et al., 2012). Flow intermittence thus increased the importance of heterotrophic processes and the contribution of the hyporheic zone to overall ecosystem processes. However, there is

scant knowledge of how interactions between ground, hyporheic, and surface waters, coupled with flow intermittence, affect the trophic ecology of IRES at higher levels of organization.

4.7.5 EFFECTS OF REGIONAL DIFFERENCES ON FOOD WEBS IN IRES

IRES vary regionally in their hydrology (Chapters 2.2 and 2.3) and biota (Chapters 4.2–4.6); IRES in arid, Mediterranean, polar, tropical, and temperate zones all have distinct climates and flow regimes. This, combined with regionally distinct biota, suggests that IRES food webs should also vary regionally. In Australia's dryland floodplain IRES, the major sources of organic material fuelling higher trophic levels shift from algae in the littoral zones of isolated waterholes during dry phases to material produced on the floodplain during, and shortly after, unpredictable floods (Bunn et al., 2006a; Balcombe et al., 2015; Fig. 4.7.7). In regions with shorter and more predictable dry phases like the Mediterranean and tropics, variation in IRES food web dynamics may be more seasonal. Evidence suggests, however, that high levels of omnivory within food webs of such regions buffers them against fluctuations in flow extremes (e.g., Mediterranean: López-Rodríguez et al., 2012; Sánchez-Carmona et al., 2012; Vannucchi et al., 2013; tropics: Leigh et al., 2010b; Pusey et al., 2010; Blanchette et al., 2014).

Notwithstanding seasonal variation in light, temperature and instream primary production, algae may be a more important basal resource in the food webs of IRES in arid, Mediterranean, and tropical regions (e.g., Álvarez and Pardo, 2009; Pettit et al., 2012; Jardine et al., 2013) than in temperate-zone IRES with well-vegetated riparian zones (e.g., Reid et al., 2008), high litter fall, and comparatively high abundance or diversity of shredders (contrast with arid-zone streams: Bunn et al., 2006b; Fig. 4.7.7). However, riparian communities and leaf-fall timing in relation to hydrological cycles influence the relative importance of autochthonous and allochthonous contributions (Section 4.7.3). In temperate Australia, algae increase in importance as a food resource for aquatic consumers during disconnected pool phases (Reid et al., 2008). In La Solana in Mediterranean Spain, shredders dominate only during periods of major leaf fall, prior to seasonal flow decline (Muñoz, 2003). The dominance of deciduous species in riparian forests of Mediterranean IRES such as La Solana likely also explains the similarities in organic matter dynamics with deciduous forest streams in temperate zones. In Fuirosos River, for example, leaf fall predominantly occurs in autumn and plays an important role in fuelling production in the following season or seasons (e.g., Dekar et al., 2009), in both this (Acuña et al., 2007) and other temperate-zone IRES. In dryland IRES, sparse riparian tree cover limits wood and leaf loads, and low numbers or absence of aquatic invertebrate shredders can limit the aquatic breakdown of leaf litter (Bunn et al., 2006b).

Regional differences in leaf-litter palatability and decomposition rates also play a role in driving regional food web differences. For example, eucalypts are common along many Australian IRES, often dominating the riparian zones of arid-zone IRES. Eucalypts produce litter with relatively low nutritional quality, low palatability to invertebrates, and slow breakdown rates (Francis and Sheldon, 2002). In a South Australian IRES, decomposition of eucalyptus leaves in isolated pools darkened waters and created oxygen-poor conditions lasting up to three months; only two species of larval caddisflies persisted under these conditions (Towns, 1991). Similar changes in water quality, litter decomposition, and invertebrate mortality are observed in IRES in regions where eucalypts are an introduced species (e.g., Canhoto et al., 2013). Considerable regional variation can thus be expected in IRES food webs based on vegetation differences alone.



FIG. 4.7.7

Vegetation in riparian zones of IRES from different regions of the world illustrating differences in density, species, structure, and complexity that likely lead to large variation in influences on food webs: (a) Macadam Creek, during its disconnected waterhole phase in the dry season of Australia's wet-dry tropics; (b) Todd River, Alice Springs in central dryland Australia; (c) Ormiston Creek, central dryland Australia; (d–f) the Santa Maria de Palautordera (d) and Sant Esteve de Palautordera (e and f) reaches of Tordera Stream (Spain) that dry every summer (f) because of water diversion; and (g) the intermittent braided channel of Emanon Creek in the tussock- and shrub-dominated Canterbury high country (~900 m asl, South Island, New Zealand).

Photos: Courtesy C. Leigh (a–c), E. García-Berthou (d), R. Merciai (e, f), and A.R. McIntosh (g).

4.7.6 STRUCTURE OF IRES FOOD WEBS

Despite regional differences in IRES hydrology and energy supply at the base of food webs, consistent patterns emerge in structural properties of food webs. The most obvious changes in drying IRES are the loss of aquatic species and associated changes in community structure; these changes are often cyclic and vary seasonally with flow (Closs and Lake, 1994; Chapters 4.4 and 4.5). This affects energy flow, patterns of food web connectivity and, ultimately, food web dimensions in aquatic ecosystems (Fig. 4.7.1), but also creates opportunities for terrestrial consumers. Changes in terrestrial food web dimensions and aquatic-terrestrial connections (Section 4.7.4) are therefore also likely to occur (Giling et al., 2015).

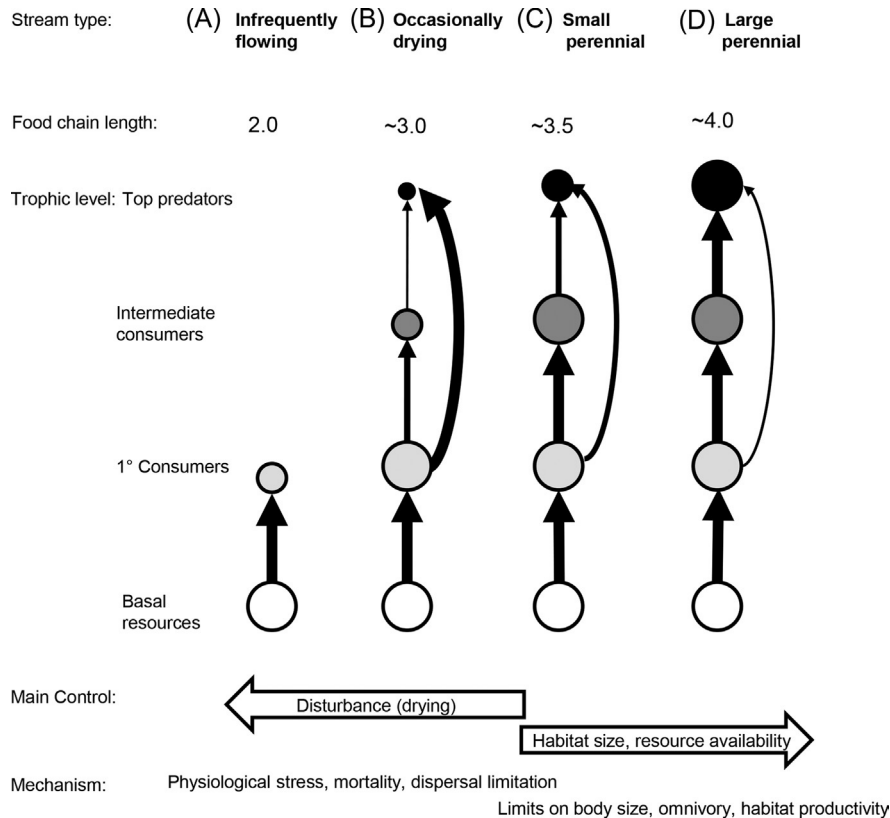
The loss of aquatic-obligate species associated with flow loss from drying invariably shrinks food web dimensions. For example, in the intermittent Lerderderg River (Victoria, Australia), where detailed food webs were constructed for pools over four time periods spanning an annual hydrological cycle, web size (S , nodes in the food web) varied seasonally from 53 to 24 (Closs and Lake, 1994). Similarly, across the longitudinal perennial-to-drying gradients of IRES in the South Island, New Zealand, S declined from 30 to 15 taxa (McHugh et al., 2015).

A major consequence of changes to web size with drying is reduced trophic area. Trophic area reflects the horizontal (trophic breadth) and vertical (trophic height) dimensions that comprise all feeding interactions and the resource use diversity within a community. Informed by stable carbon-13 and nitrogen-15 isotope ratios, which respectively reflect the horizontal and vertical dimensions (Layman et al., 2007), trophic area signals energy flow through food webs. McHugh et al. (2015) observed a 20-fold decrease in trophic area, primarily driven by reduced trophic height, as habitat size decreased along drying continua. Changes in horizontal food web dimensions also occur, potentially reflecting altered use of basal resources (Section 4.7.4). The dominant change, however, is in trophic height, rather than in the breadth of resources used, perhaps because work to date could not fully resolve basal resource pools.

Reduced trophic height signals a reduction in aquatic food chain length (FCL). FCL reductions have been observed in many drying streams (Fig. 4.7.8), suggesting a consistent response of food webs to drying (Closs and Lake, 1994; Sabo et al., 2010; Woodward et al., 2012; McHugh et al., 2015). In a New Zealand study, FCL differed by approximately two trophic levels between perennial and drying sites. Similarly, FCL was approximately one trophic level longer in perennial rivers compared to IRES in the United States (Sabo et al., 2010). This reduction in FCL is more dramatic than those driven by increasing flood-related disturbance and is also larger than the increase in trophic levels seen along a gradient of increasing stream size (McHugh et al., 2010; Sabo et al., 2010). Interestingly, although seasonal increases in water supply in dynamic systems with extensive changes in the dimensions of aquatic habitat, such as the floodplain IRES of central and northern Australia, can increase consumer dietary breadth (Balcombe et al., 2005), they may not translate into major shifts in FCL (Warfe et al., 2013).

Drying-linked reductions in FCL can be characterized as trophic collapse and are typically associated with the loss of large-bodied top predators, typically fish or predatory invertebrates. Sabo et al. (2010) found top predators in food webs to differ between IRES and perennial streams; in perennial streams, fish were often large and piscivorous whereas top predators in IRES were either invertebrates or smaller insectivorous fish.

Importantly, the shape of the relationship between flow and FCL may be nonlinear in IRES, suggesting there are size thresholds associated with trophic collapse (McHugh et al., 2015). Declines in

**FIG. 4.7.8**

Mechanisms influencing food chain length (FCL) in New Zealand IRES, relative to perennially flowing habitats. *Circles* represent food chains of consumers of varying abundance (*circle size*) feeding at progressively higher trophic levels, with arrows indicating relative energy flow (*arrow size*). At the extreme dry end of the IRES spectrum (A), food chains are short because few aquatic taxa can withstand complete desiccation/infrequent rewetting and/or cannot access these sites before they dry again. At the wet end of the IRES spectrum (B), frequent drying disturbance, extreme habitat compression, and/or stress reduce the diversity and size of taxa present, and disproportionately affect higher trophic level taxa. Additionally, because these sites are inherently more variable than perennial sites, top-level taxa are expected to tend toward more trophic omnivory where they also feed on primary consumers more than in more stable, perennial sites resulting in net food chain lengths close to three trophic levels. Food chain lengths in perennial streams (C, D) are expected to vary as a function of habitat size (discharge), resource availability, and disturbance resulting from extreme high flows, with the greatest vertical structuring/FCL observed in large stable systems where most top-level consumers are feeding high up the food web, resulting in food chain lengths closer to four trophic levels.

Based on McHugh, P., Thompson, R.M., Greig, H.S., Warburton, H.J., McIntosh, A.R., 2015. Habitat size influences food web structure in drying streams. *Ecography* 38, 700–712.

overall habitat size that limit body sizes for high-level consumers (e.g., fish, Walters and Post, 2008) are likely an important driver here, just as they are in perennial streams (McHugh et al., 2010; Sabo et al., 2010). However, in IRES declines are also likely linked to drying-associated stress (HersHKovitz and Gasith, 2013) because FCL-habitat size scaling is steeper in IRES than in perennial streams. McHugh et al. (2015) found FCL to be 0.25–0.50 less in IRES than in perennial reaches of equivalent size. Similarly, Sabo et al. (2010) found that FCL in drying streams in the United States was around two-thirds less compared to perennial streams. Thus, flow cessation initially constrains available space, reducing the size of organisms present (Walters and Post, 2008; Jellyman et al., 2014), and subsequently FCL (e.g., Romanuk et al., 2011). However, severe habitat shrinkage also subjects organisms to harsh conditions associated with, for example, warmer temperatures, reduced oxygen, and lower total resource abundances.

The harshness of intermittence is reflected in decreased energy intake and condition for fish as flow declines (Closs, 1994; Mas-Martí et al., 2010). Larger, longer-lived predatory taxa are often the slowest to recolonize streams upon flow resumption (e.g., Closs and Lake, 1994), consistent with theoretical models linking disturbance and demography to food web structure (e.g., Pimm and Lawton, 1977), although population resistance to drying-related disturbance depends heavily on the magnitude of stress and minimum habitat size (White et al., 2016). Thus, shrinking habitat size and deteriorating conditions force predatory fish to leave or suffer, meaning that top predator loss is an important proximate mechanism (*sensu* Post and Takimoto, 2007) of food web change in IRES. This phenomenon may result from natural or human-driven dewatering associated with abstraction (Chapter 5.1) and has implications for determining environmental flow regimes if damaging effects are to be avoided.

Flow intermittence also affects FCL by reducing the abundance of organisms living at intermediate trophic levels, like predatory invertebrates, or the degree to which top predators feed across trophic levels (i.e., insertion and omnivory, Post and Takimoto, 2007). In contrast to fish, which often respond predictably to low flows and drying (Walters, 2016), invertebrate responses are highly variable (Dewson et al., 2007). Obligate-aquatic invertebrates may respond similarly to fish (i.e., they are quickly lost and slow to recolonize; Closs and Lake, 1994). However, owing to a greater diversity of invertebrate adaptations (Chapters 4.3 and 4.8) for withstanding flow extremes, few generalities can be made regarding which taxa persist at low flow states (Walters, 2011).

While comparative studies examining the effects of extreme flow reduction reveal effects ranging from strong to weak on FCL in aquatic food webs, experiments reveal the mechanisms underlying changes. In general, S , body-size range and overall food web connections decreased in dewatered stream mesocosms in Dorset, UK (Woodward et al., 2012; Ledger et al., 2013), in line with the comparative studies mentioned earlier. However, some network-level properties were unaffected, including connectance, linkage density (i.e., links/species), and interaction diversity. Similarly, Walters and Post (2008, 2011) found that FCL resisted change in response to experimental dewatering, although they noted a response in body size and invertebrate communities to treatments. Thus, while some trophic dimensions of food webs respond considerably to habitat contraction, some structural properties of food webs can be maintained in the face of extreme disturbance.

Given the patterns described earlier, it is important to distinguish between disassembled food webs, those in the process of disassembly and those undergoing repeated disassembly. Webs observed in IRES are often a snapshot, at some stage of disassembly. Their configurations may thus be unstable, perhaps top-heavy or with a high predator-prey ratio (Bogan and Lytle, 2007; McHugh et al., 2015). This could make food webs of drying IRES vulnerable to strong trophic cascades (Bogan and Lytle,

2011). However, the Dorset food webs were exposed to repeated disturbance and had a stable structure, albeit with S considerably reduced. There may be mechanisms which, over time, work to reduce the destabilizing effects of drying, yielding potentially “disturbance-adapted” food webs. Closs and Lake (1994) argued that weak top-down trophic interactions, a heavy reliance on detrital food resources, and cross-ecosystem subsidies were important contributors to the stability of the Lerderderg River food web. Overall, these stabilizing and destabilizing forces require more in-depth study and are particularly relevant in the management of IRES because they will be important in maintaining ecosystem functions in IRES.

Contractions in aquatic food web dimensions offer potential for a corresponding increase in terrestrial food web dimensions with alterations in energy flow toward terrestrial top predators like birds, bats, and spiders (Section 4.7.4). Drying and the subsequent entrapment of aquatic organisms can provide a feeding bonanza for terrestrial predators and decomposers (e.g., Dekar and Magoulick, 2013). Understanding the dynamic temporal changes associated with drying, where potentially large amounts of carbon are transferred across ecosystem boundaries in very short time periods, raises a challenge for food web ecologists. Closs and Lake (1994) regarded their Lerderderg River food web as a “sub-web contained within the overall Lerderderg valley food web” (p. 17), and suggested that FCL would inevitably be longer if insectivorous birds were considered.

4.7.7 TROPHIC INTERACTIONS IN IRES IN THE FUTURE

Global change, especially land-use change and climate warming, is expected to increase the spatial and temporal fragmentation of flowing waters (Datry et al., 2014; Chapter 5.1). Given that many of the changes associated with drying increase interaction strengths and produce conditions which potentially destabilize food webs, IRES systems are inherently vulnerable. Some evidence suggests systems adapted to these conditions will remain resilient, but even these systems respond to extended drying, such as during Australia’s “millennium drought” (Lake, 2011). Understanding controls on system stability will be particularly important if deleterious consequences are to be avoided, suggesting this is an important area for future research. Moreover, to avoid extinctions, the plight of systems which appear to lack resistance and resilience to drying, such as those associated with the giant water bugs (Section 4.7.3), must be carefully considered.

A critical feature of IRES food webs is their ability to perform ecosystem functions and provide ecosystem services. Although we have a preliminary understanding of food web structure, we have a relatively poor understanding of their functioning. Key aspects include their role in nutrient cycling (Chapter 3.2), the cross-ecosystem energy transfer, and their potential to sustain unique elements of biodiversity.

Large-bodied organisms, including many top predators, are an especially vulnerable group supported by IRES. Studies in both experimental and natural systems indicate large taxa are disproportionately vulnerable to drying (Woodward et al., 2012; Ledger et al., 2013; McHugh et al., 2015). Moreover, because aquatic ectotherms are particularly sensitive to temperature changes, flow loss effects on temperature combined with negative temperature-body size covariance (Daufresne et al., 2009) suggest large bodied taxa in IRES will be further impacted by global climate warming. Thus, maintaining refugia for these organisms will be critical (Magoulick and Kobza, 2003), especially given their potential role in maintaining system stability.

The dynamic nature of IRES habitats and food webs also presents a ripe research opportunity to more effectively incorporate temporal variation into food web theory and network science, as called for in recent reviews (Thompson et al., 2012). Moreover, the spatial interactions of IRES, which occur in multiple dimensions, offer further opportunities to enhance understanding of the changing spatial dimensions of food webs. Answering critical questions concerning trophic interactions in IRES will thus enhance knowledge of more general processes in food web ecology.

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FURTHER READING

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